

EKG Design Project

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Design/Solution:

The design utilizes 3 stages for improving signal integrity. The first stage used an instrumentation amplifier to amplify the input signal by a factor of 7, and the output was then filtered with a simple RC circuit. After filtering, a second-stage amplifier with a gain of 101 V/V was applied to bring the total gain to approximately 707 V/V. Then, to keep the signal within the PIC24 ADC range, a summing amplifier was used to add a 1.5V DC offset to the signal. Lastly, a unity-gain inverting amplifier was used to feed the common-mode signal back to the body, reducing noise in the input signal.

The common mode signal represents the voltage pulse delivered by the human heart pulse detected by electrodes. This signal is divided into two differential signals. The two differential signals are amplified by an instrumentation amplifier (Chip used: INA333) with a gain set based on the R_G terminal voltage drop, which is fed back to the Right-Leg Drive network. Furthermore, the output of the instrumentation amplifier (Chip used: INA333) would pass through an RC high-pass filter with a corner frequency of approximately .026Hz, to allow the heartbeat signal to pass through while blocking lower-frequency drift. The final stage of our analog design amplifies the signal (Chip used: OPA333) for reading by the PIC24 ADC, which displays the heartbeat waveform in real-time using a Pyplotter to demonstrate the achieved ECG heartbeat. Moreover, an applied DC offset is added to the ADC input using the OPA33 as a summing amplifier. Our team has designed an analog circuit shown in Figure 1-1 below.

The circuit was first assembled on a breadboard to confirm that the analog design worked, as it was expected to produce a reliable reading of the human pulse. Once a member connected their arms and right leg to the electrodes, we were able to track the human pulse waveform using an oscilloscope and a power supply (which was later replaced with a battery). Initially, some noise was discovered that made the human pulse nearly impossible to detect. A solution to this problem was to change our RC values on the breadboard to smooth the signal noise, which worked. Once we transitioned to the PIC24 and a UART connection, the software used to track the human pulse ADC signal for the ECG in real-time was Pyplotter, which, at this point in troubleshooting, exhibited constant 60Hz noise. A notch filter was added at the ADC input to reduce the 60Hz noise.

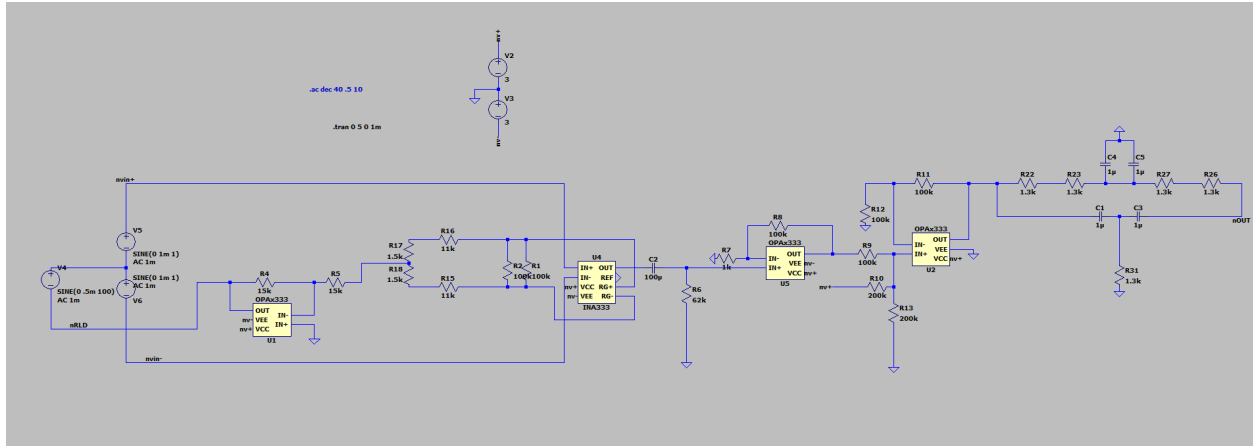


Figure 1-1: Analog Design

Figure 1-2, shown below, shows the final waveform obtained from LTSPICE simulations. A DC offset of 1.65 V was obtained from the analog design at the nout node using LTSpice. The overall gain of our circuit, which is about 707 for this 1Hz signal, acting on a 2mV_{pp} input, puts the output signal at almost the full range of the ADC interface. 2mV was in reality, larger than any heartbeat signal we observed which guaranteed that the EKG output would always stay in the ADC range with a peak to peak amplitude of 1-2V_{pp}. The time period of this voltage signal for our output signal V(nout) is $T_{\text{period}} = 1\text{s}$, so that our frequency is $f_{\text{period}} = 1\text{Hz}$. This corresponds to 60bpm which is a common heart rate.

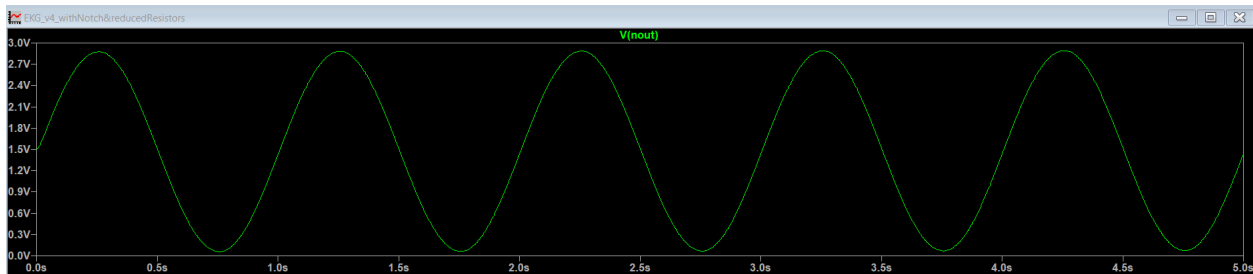


Figure 1-2: Simulated Final Stage Output Waveform to interact with ADC of the PIC24

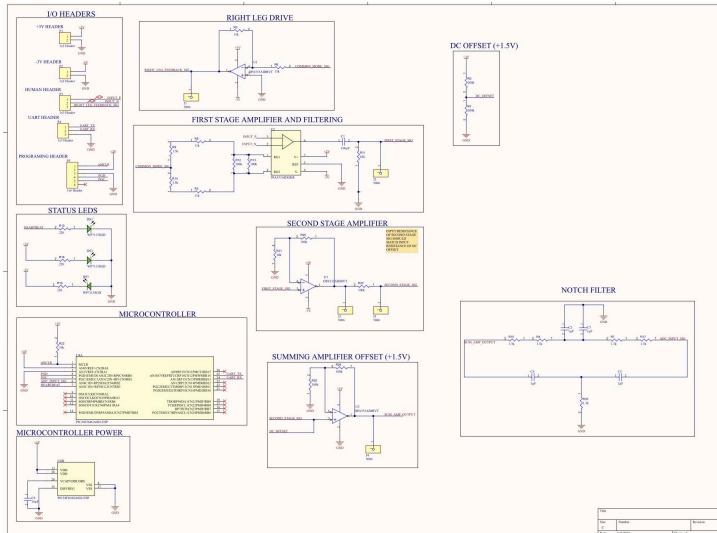


Figure 1-3: Final Altium Schematic

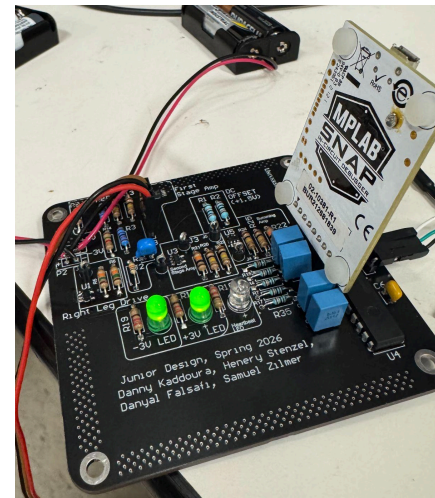


Figure 1-4: Final PCB

For the first order of the PCB design, an issue was found before any testing: we accidentally made the PCB for an OPA333 using a different package and thus a different pinout than the OPA333 we ordered. This meant we could not test the PCB without ordering the other OPA333. We also did not include a notch filter for 60Hz interference. Moreover, this required us to add a notch filter and reorder a few line traces on our PCB. The notch filter eliminated most of the noise, giving us the heartbeat reading our ECG was designed to show on both a digital display and an analog waveform. Additionally, for the second PCB, specific resistor values were ordered from DigiKey, which led to reducing parallel/series connections.

Conclusion:

Overall, our project closely met expectations. Our project was assembled on a custom PCB and smaller than 6". Our analog circuit was powered with dual 3V battery power supplies. We had an analog output with an amplitude of $1-2V_{pp}$ depending on the person attached to the electrodes, with a DC offset of approximately 1.65V. The heartbeat was well-defined and filtered in the output of the pyplotter. Subsequent digital filtering was thus proved unnecessary. The quality of the output was good. When the subject moved, there was minimal drift in the output waveform, and the 60Hz noise was filtered by the notch filter. The main source of difficulty was the 60Hz noise consistently observed in the output waveform; however, after discussion with other groups, it was determined that implementing a notch filter before the final output would resolve the issue, yielding a better-defined output heartbeat waveform.